Colligative properties of electrolyte

1. If O.P. of 1 *M* of the following in water can be measured, which one will show the maximum O.P.
   (a) $AgNO_3$ (b) $MgCl_2$
   (c) $(NH_4)_3PO_4$ (d) Na_2SO_4

2. Which of the following solution in water possesses the lowest vapour pressure
   (a) $0.1(M)NaCl$
   (b) $0.1(N)BaCl_2$
   (c) $0.1(M)KCl$
   (d) None of these

3. Which of the following solutions in water will have the lowest vapour pressure
   (a) 0.1 *M*, *NaCl*
   (b) 0.1 *M*, Sucrose
   (c) 0.1 *M*, $BaCl_2$
   (d) 0.1 *M* Na_3PO_4

4. The vapour pressure will be lowest for
   (a) 0.1 M sugar solution
   (b) 0.1 M *KCl* solution
   (c) 0.1 M$Cu(NO_3)_2$solution
   (d) 0.1 M $AgNO_3$ solution

5. Osmotic pressure of 0.1 M solution of $NaCl$ and Na_2SO_4 will be
   (a) Same
   (b) Osmotic pressure of $NaCl$ solution will be more than Na_2SO_4 solution
   (c) Osmotic pressure of Na_2SO_4 solution will be more than $NaCl$
   (d) Osmotic pressure of $NaSO_4$ will be less than that of $NaCl$ solution

6. Which of the following solutions has highest osmotic pressure
   (a) $1MNaCl$
   (b) 1 M urea
   (c) 1 M sucrose
   (d) 1 M glucose

7. Which one has the highest osmotic pressure
   (a) $M/10HCl$
   (b) $M/10$ urea
   (c) $M/10BaCl_2$
   (d) $M/10$ glucose

8. In equimolar solution of glucose, $NaCl$ and $BaCl_2$, the order of osmotic pressure is as follow
   (a) Glucose $> NaCl > BaCl_2$
   (b) $NaCl > BaCl_2 >$ Glucose
   (c) $BaCl_2 > NaCl >$ Glucose
   (d) Glucose $> BaCl_2 > NaCl$

9. The osmotic pressure of which solution is maximum (consider that

deci-molar solution of each 90% dissociated)
(a) Aluminium sulphate
(b) Barium chloride
(c) Sodium sulphate
(d) A mixture of equal volumes of (b) and (c)

10. At 25^oC , the highest osmotic pressure is exhibited by $0.1M$ solution of
(a) $CaCl_2$ (b) KCl
(c) Glucose (d) Urea

11. Which of the following will have the highest boiling point at 1 *atm* pressure
(a) $0.1MNaCl$
(b) $0.1M$sucrose
(c) $0.1MBaCl_2$
(d) $0.1M$ glucose

12. Which one of the following would produce maximum elevation in boiling point
(a) 0.1 M glucose
(b) 0.2 M sucrose
(c) 0.1 M barium chloride
(d) 0.1 M magnesium sulphate

13. Which of the following solutions will have the highest boiling point
(a) 1% glucose
(b) 1% sucrose
(c) 1% $NaCl$
(d) 1% $CaCl_2$

14. Which one of the following aqueous solutions will exhibit highest boiling point
(a) 0.015 M urea
(b) $0.01MKNO_3$
(c) $0.01MNa_2SO_4$
(d) $0.015M$glucose

15. Which of the following aqueous solutions containing 10 gm of solute in each case has highest B.P.
(a) $NaCl$solution
(b) KCl solution
(c) Sugar solution
(d) Glucose solution

16. 0.01 molar solutions of glucose, phenol and potassium chloride were prepared in water. The boiling points of
(a) Glucose solution = Phenol solution = Potassium chloride solution
(b) Potassium chloride solution > Glucose solution > Phenol solution
(c) Phenol solution > Potassium chloride solution > Glucose solution

(d) Potassium chloride solution > Phenol solution > Glucose solution

17. Which one has the highest boiling point

(a) $0.1 N Na_2SO_4$

(b) $0.1 N MgSO_4$

(c) $0.1 M Al_2(SO_4)_3$

(d) $0.1 M BaSO_4$

18. Which of the following solutions boils at the highest temperature

(a) 0.1 *M* glucose

(b) 0.1 *M NaCl*

(c) 0.1 *M* $BaCl_2$

(d) 0.1 *M* Urea

19. $0.01 M$ solution each of urea, common salt and Na_2SO_4 are taken, the ratio of depression of freezing point is

(a) 1 : 1 : 1 (b) 1 : 2 : 1

(c) 1 : 2 : 3 (d) 2 : 2 : 3

20. Which has the minimum freezing point

(a) One molal $NaCl$ solution

(b) One molal KCl solution

(c) One molal $CaCl_2$ solution

(d) One molal urea solution